02-719 Week 7

Genomics and Epigenetics of the Brain

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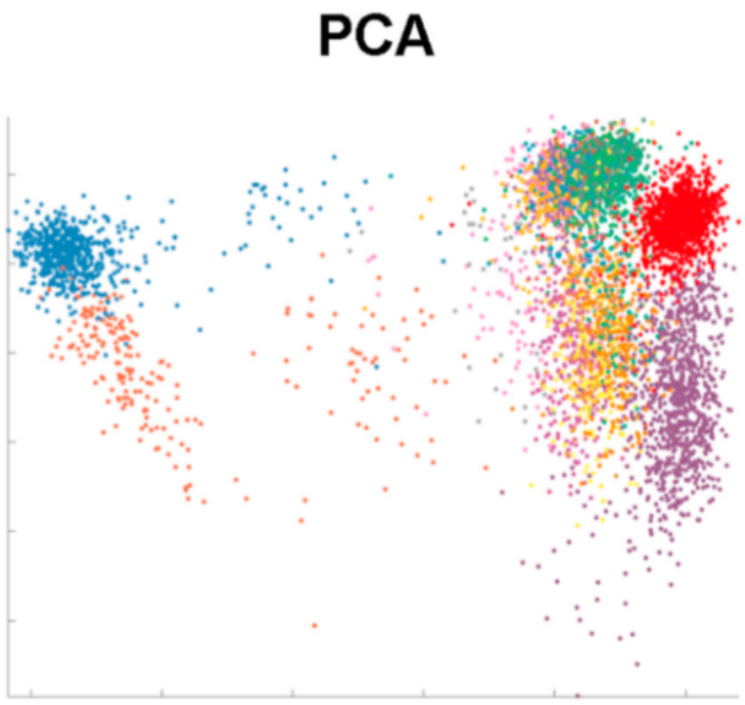
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t-SNE

First, let's consider a PCA of a cell dataset.

Recall PCA is a dimensionality reduction technique.

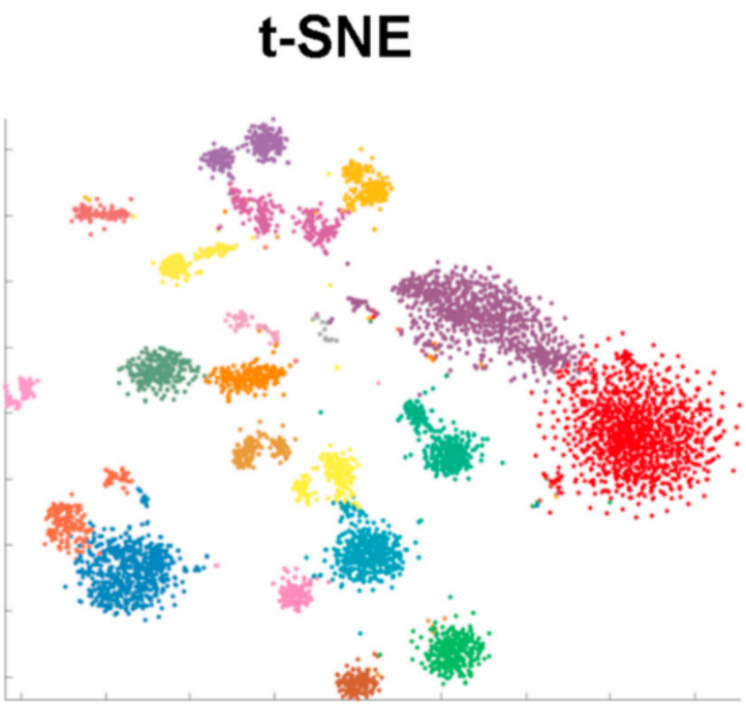
- Linear
- Flavor of Matrix Factorization
- Maximizes variance explained by components



Here, each point is a cell, and each color is a cell type in the retina.

- Why doesn't this look good?
- Sometimes, the data isn't linearly separable!

Now, let's try the nonlinear method, t-SNE:



- This is much better!
- Unlike PCA, t-SNE is able to capture nonlinear dynamics. (These are very prevalent in snRNA-req data).
- Each PC in snRNA-seq data usually captures $< 1\%$ of the total variance
- PCA focuses on global variance so the first few components may not include information on more granular structure

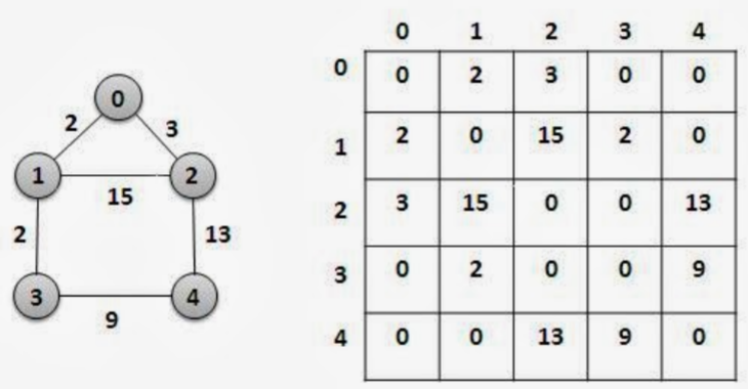
How can similarity be defined?

It is usually defined based on the PCA metagene matrix we made

- Cosine distance/similarity
- Euclidean distance can be turned into similarity if taken through a gaussian kernel
- Number of shared nearest neighbors can define similarity

The similarity matrix can be viewed as a graph

- Nodes are cells
- Edges are the similarity between 2 cells.



	0	1	2	3	4
0	0	2	3	0	0
1	2	0	15	2	0
2	3	15	0	0	13
3	0	2	0	0	9
4	0	0	13	9	0

TSNE-T distributed stochastic neighbor embedding

- Tries to replicate the similarity found in the similarity matrix in two dimensions
- Actually a probabilistic approach
  - Given a distance between two cells it creates a probability that they should be distance X away from each other in the 2d space
- This probability is based on the t-distribution with defined parameter perplexity

TSNE - How perplexity can change the visualization

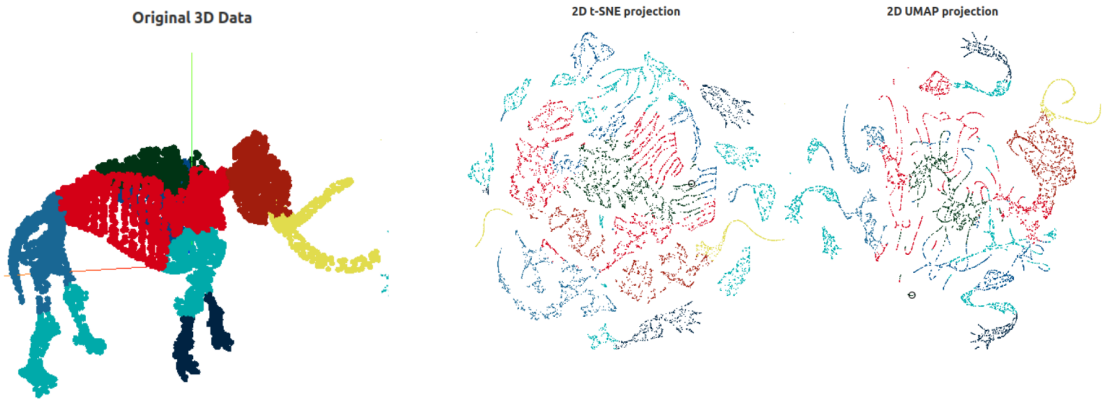


UMAP

UMAP is an alternative to TSNE.

- Instead of the probabilistic distance between cells this focuses on the topology or graph structure of that similarity matrix and tries to maintain it in the 2d projection
- Two parameters:
  - `min_dist`: the minimum distance considered meaningful
  - `N_neighbors`: how far out in the graph does structure matter?

UMAP on a 3D Mammoth



Comparison of Dimensionality Reduction Methods

PCA	UMAP / TSNE
Linear	Non-linear
Dimensions are explainable by gene embeddings	Dimensions have no intrinsic meaning
Deterministic (will get the same dimensions every time)	Non-deterministic
Captures small amount of variance per dimension	2-3 dimensions can give a good look at the variability across your dataset

Note that **UMAP and t-SNE are *approximations* of the similarity matrix**.

- They should be used for visualization, data exploration, and hypothesis generation
- They should not be used to draw statistical conclusions about your data

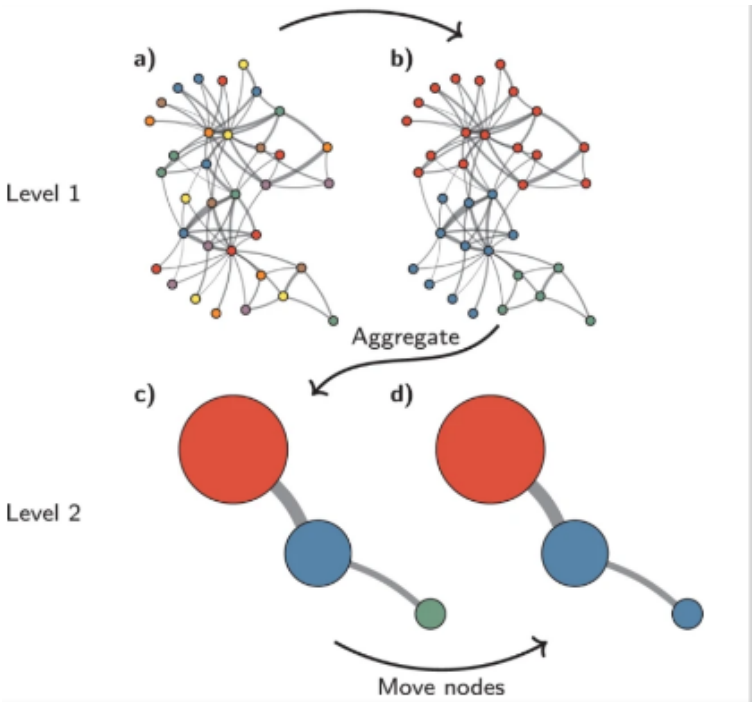
Clustering

Louvain Community Detection

- Community detection is a type of clustering performed on a similarity graph
- Find communities (clusters) of tightly knit nodes
- Optimizes modularity

Louvain Algorithm Steps

- The Louvain algorithm consists of 2 repeating steps:
  - Step 1 moves nodes to the optimal community
  - Step 2 aggregates all nodes in a community
- Each repetition of these steps is called a level
- The steps are repeated until convergence (nothing changes between levels)



Step 0: Initialization

Each vertex v_i has its own community c_i.

Step 1

Step 1 can be divided into passes. In each pass, the following algorithm is used:

- For each vertex v_i,
  - For each community c_j
    - If adding vertex v_i to community c_j increases the modularity of the similarity graph, add v to c.

Step 2

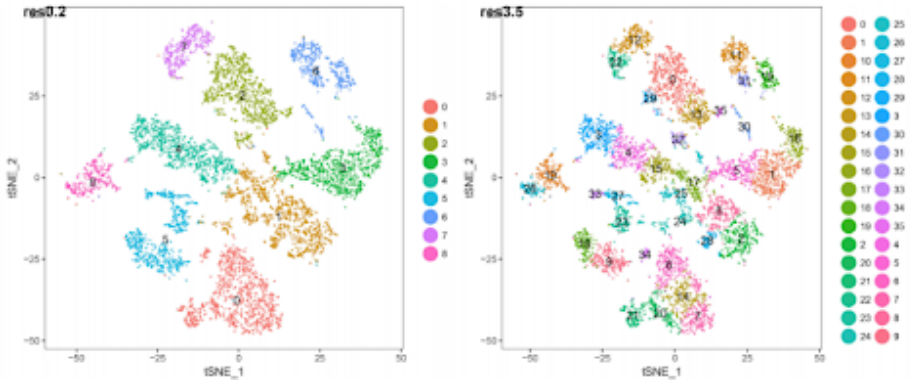
Step 2 is the merge step - all the vertices in each community are merged into a single vertex. The edges between communities are the sum of edges between the vertices in the communities. After this, repeat the process until convergence.

Adding a Resolution Parameter

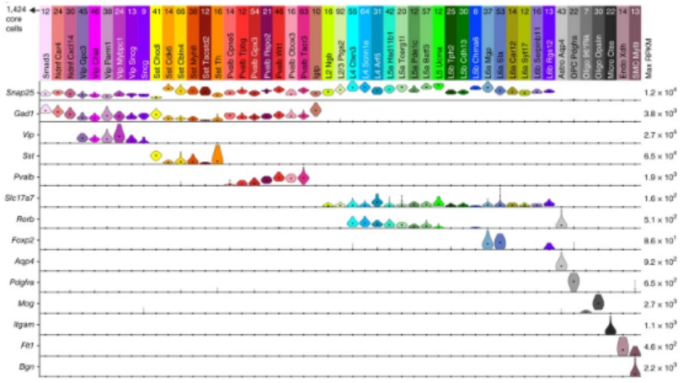
- You will likely want to cluster your single cell data set at different resolutions.
- To address this we can add a resolution parameter to weight p_{ij}
- The higher the resolution the more fine grained the clusters will become

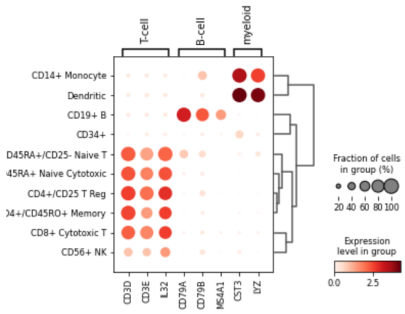
$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{i,j} - r \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

Effects of the Resolution Parameter



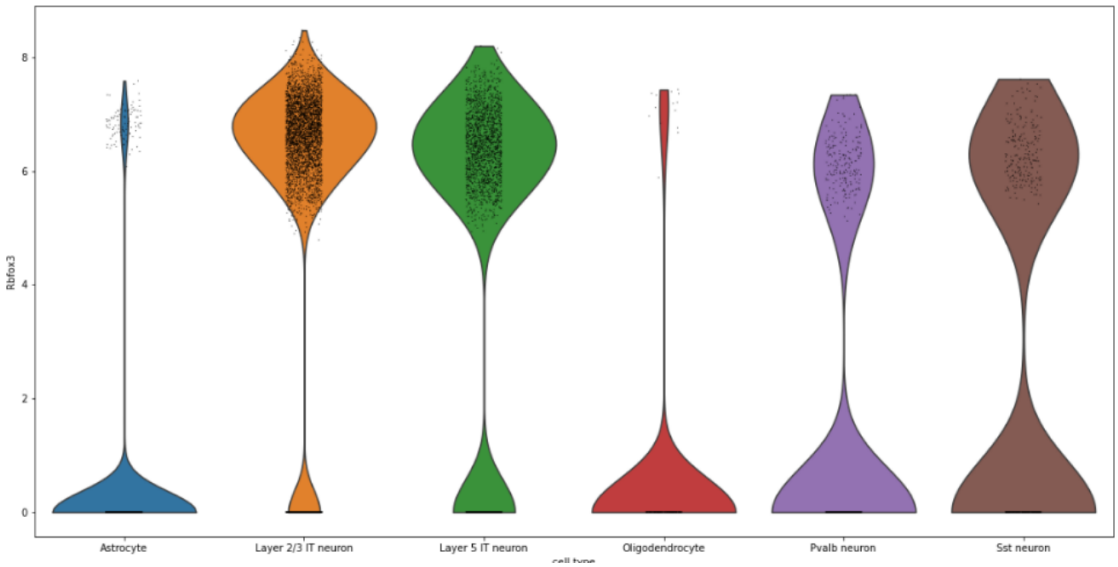
Marker Gene Plots



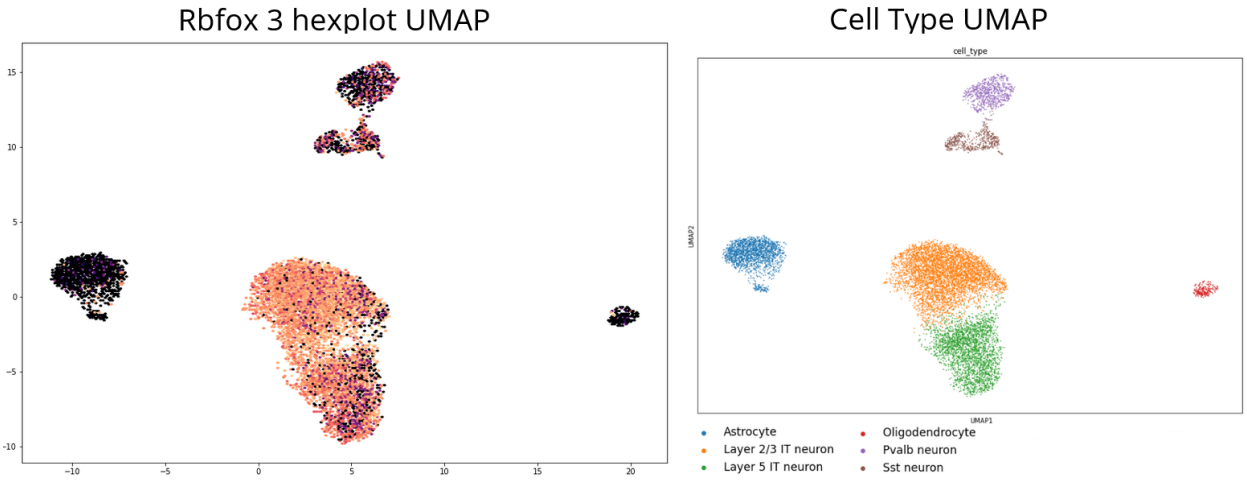


Rbfox 3 on a Violin Plot Mouse Motor Cortex Cells

Let's use an example: Rbfox3 is a marker gene for neurons.



Rbfox3 on a UMAP of Mouse Motor Cortex Cells



Marker Genes are Relative

- A marker gene allows a cell type to be identifiable compared to every other cell type in the population you are analyzing
- If you have the same cell type with a different population in your analysis, you may get different marker genes!

Differential Expression Tests

- A differential expression statistical test is used to see which genes differ in their expression distribution between 2 populations
- It has the following null hypothesis for each gene X: the expression of gene X was drawn from the same distribution for both population A and B.
- To perform these tests we must first normalize the expression values

Wilcoxon-Rank Sum Test

- The T-test is a *parametric* test, which means it makes assumptions about the types of distributions that the statistic is calculated from. Namely, the t-test assumption is the difference between expression distributions is gaussian.
- The Wilcoxon-rank sum test makes no such assumptions. It is *non-parametric*.
- Instead of using mean or variance measurement it uses a difference in ranks for its statistic.

More Complex Experimental Designs

- For these marker genes we only looked at the difference between two groups.
- Why may this be too simplistic?
  - We might want to take into account other factors such as: sequencing batch, sex of sample, age/time point at which the sample was taken, multiple treatment conditions, disease progression, etc.

Linear Models

Linear models allow you to take into account multiple factors when calculating differential expression.

- Rbfox3 expression $= \beta_0 + \beta_1 + \beta_2 + \beta_3$, where $\beta_1 :=$ batch, $\beta_2 :=$ sex, $\beta_3 :=$ is neuron.

In this case our statistic will be β_3 and our p-value will be calculated based off of it. There are many packages to use linear models for RNA-seq data, for single cell recommend limma-trend.